

# Clothing Brand E-Commerce Platform — Software Requirements Specification (SRS) v1.1

**Author:** [Your Name / Team]

**Owner:** [Client Name / Company]

**Date:** [Insert Date]

**Status:** Updated with Personalization Requirements

## 0. Document Control

Purpose: Define complete functional and non-functional requirements for an e-commerce website for a clothing brand. Serves as the authoritative reference for product, engineering, design, and operations teams.

Audience: Product Owner, Developers, Designers, QA, Operations, Finance, Management.

Change Process: All updates tracked in version control. Major changes require product owner approval.

## 1. Product Overview

A responsive web-based platform where customers can browse, customize, and purchase clothing items. The site provides information about the company, its training programs, and enables administrative management of products, orders, and users. **1.1 Goals & Success Metrics**

- G1: Provide a smooth and engaging online shopping experience.
- G2: Allow customers to customize products and preview designs.
- G3: Enable administrators to easily manage products, orders, and training programs.
- G4: Provide company management with actionable sales reports and analytics.

### KPIs:

- Number of monthly active users
- Conversion rate (visitors → buyers)
- Average order value
- Number of custom product orders
- Sales per category and per product **1.2 Out of Scope (v1)**
- AI-driven design recommendations
- Multi-vendor marketplace functionality
- Cross-border inventory synchronization

## 5. Functional Requirements

### Authentication & Authorization:

Roles: Customer, Admin, Boss. Secure login via email/password. Role-based access control using Laravel middleware.

### Product & Catalog Management:

CRUD for products and categories. Upload multiple images. Manage stock.

### Personalization Engine (Advanced):

- Multi-step configurator similar to automotive personalization tools.
- Users choose base product (e.g., T-shirt, hoodie) and customize size, color, and style.
- Live price updates as users add features or upload artwork.
- Interactive preview:
  - 2D mockup with selectable sides (front, back, sleeves).
  - Option for 360°/multi-angle view using pre-rendered images or 3D model.
- Users can upload logos, add text, move/scale/rotate elements within safe print areas.
- Price calculation based on customization complexity (e.g., number of sides, print area size).
- Save & Share feature:
  - Logged-in users can save a build for later editing.
  - Public share link for showing friends or collaborating on designs.
- Production-ready file generation upon checkout (high-res PNG/SVG with bleed/crop).
- Moderation and validation:
  - Artwork resolution check (warn if below required DPI).

- Content moderation pipeline to block inappropriate uploads.

**Shopping Cart & Checkout:**

Add/remove/update items. Secure checkout with Stripe, PayPal, or Mobile Money.

**Training Program Management:**

Create/update programs. Accept registrations and payments. Track participants.

**Reporting & Analytics:**

Sales by product/category. Revenue over time. Most customized items. Export PDF/Excel.

## 6. Non-Functional Requirements

- Performance: Page loads < 3 seconds
- Security: Encrypted passwords, HTTPS, PCI-compliant payments
- Availability: 99.5% uptime target
- Scalability: Support 10,000 concurrent users
- Responsiveness: Works on mobile, tablet, desktop

## 7. Compliance & Trust

- Data privacy policy
- Secure handling of payment data
- GDPR-style consent for users

## 8. Operations & Reporting

- Finance reports (sales, revenue)
- Inventory tracking reports
- User activity metrics

## 9. Deployment & Maintenance

- Dockerized Laravel + React stack
- CI/CD pipeline with automated testing
- Production monitoring and logging

## 11. Closing Notes

This document defines the foundational requirements for building the clothing brand's e-commerce platform. Updates will be made iteratively based on stakeholder feedback and testing.